Theory of Universal Origins (TUO): A Pre-Planck Cosmogenesis Framework

Deriving the Hot Big Bang Initial Conditions from a Global Zero-Sum Constraint

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Abstract

The Theory of Universal Origins (TUO) proposes that the observable universe arose from a zero-sum rearrangement of a pre-existing flat Minkowski vacuum. Two axioms suffice: (I) the pre-emergence background is flat (3+1)-dimensional Minkowski spacetime, and (II) the global density operator $\hat{\rho}(t)$ satisfies $\mathrm{Tr}[\hat{\rho}(t)\,\hat{Q}_k] = 0$ for all conserved charges $\hat{Q}_k$ and all times t.

From these axioms, combined with the Heisenberg uncertainty principle and Standard Model particle content ($g_* = 106.75$ relativistic degrees of freedom), TUO *derives* the initial conditions that standard Big Bang cosmology assumes:

- Equation of state $w = \frac{1}{3}$ (radiation dominated) — derived from $v = c$ at emergence, not assumed.
- Hubble parameter $H(t_{\mathrm{Pl}}) = 1/(2t_{\mathrm{Pl}})$ — derived from wavepacket spreading, exactly matching the Friedmann prediction.
- Spatial flatness $\Omega = 1$ exactly — from Axiom I.
- Energy per Planck cell $E_{\mathrm{cell}} = (g_*/2)E_{\mathrm{Pl}} = 1.04 \times 10^{11}$ J, within a factor of 1.52 of the standard cosmological back-extrapolation.

The expansion law $V(t) = \frac{4\pi}{3}\ell_{\mathrm{Pl}}^3\left[1 + (ct/\ell_{\mathrm{Pl}})^2\right]^{3/2}$ is derived from quantum wavepacket spreading and carries a new measurable correction $\Delta V/V = 3(\ell_{\mathrm{Pl}}/ct)^2$, order-unity at t_{Pl} and negligible at all observable epochs. A path-integral saddle-point argument shows that simultaneous emergence of all $\sim 10^{93}$ Planck cells in the observable universe is exponentially preferred over any localised fluctuation, providing a mechanism for CMB uniformity that requires no inflaton field. Every fundamental equation in physics is shown to be an equality as a consequence of Axiom II: the Einstein field equations, Schrödinger/Dirac equations, and quantum entanglement (EPR) all express the zero-sum constraint in their respective mathematical languages.

Contents

1 Introduction **2**

2 Axioms **2**

3 Standard Model Content and Zero-Sum Verification **3**
3.1 Particle content . . . 3
3.2 Per-generation zero-sum . . . 3

4 Energy Prediction **4**
4.1 Heisenberg minimum energy at $t = t_{\mathrm{Pl}}$ . . . 4
4.2 Corrected formula using g_* . . . 4
4.3 Comparison with standard cosmology . . . 5

5 Stability of Matter-Only Configurations **5**

6 Path-Integral Proof of Simultaneous Emergence **6**
6.1 Setup . . . 6
6.2 Lagrange multiplier representation . . . 6
6.3 Saddle-point analysis . . . 6

7 Expansion Law from Quantum Wavepacket Spreading **7**
7.1 Initial wavepacket . . . 7
7.2 Time-evolved wavepacket . . . 8
7.3 Hubble parameter . . . 8
7.4 Sub-luminal expansion . . . 9

8 Junction Conditions at $t = t_{\mathrm{Pl}}$ **9**

9 Thermal History and BBN **10**
9.1 BBN yields . . . 10

10 Physical Equalities as Consequences of Zero-Sum **10**
10.1 Einstein field equations . . . 10
10.2 Schrödinger and Dirac equations . . . 11
10.3 EPR correlations and quantum entanglement . . . 11
10.4 Noether's theorem — inverted . . . 11

11 Quantitative Predictions **11**

12 Discussion and Open Problems **11**
12.1 Summary of resolved issues . . . 12
12.2 Open problems . . . 13
12.3 Relation to other approaches . . . 13

13 Conclusion **14**

1 Introduction

Standard Big Bang cosmology is extraordinarily successful. It predicts the cosmic microwave background (CMB) temperature, the primordial element abundances from Big Bang nucleosynthesis (BBN), and the large-scale structure of the universe to sub-percent precision [1, 2]. Yet it provides no answer to the most basic question: *why did the universe begin with those specific initial conditions?* The flatness of space ($\Omega = 1$), the radiation-dominated equation of state ($w = \frac{1}{3}$), the homogeneity of the CMB to one part in 10^5 across causally disconnected regions — all of these are inputs to the standard model, not outputs.

Inflationary cosmology partially addresses this by postulating an epoch of accelerated expansion driven by a scalar inflaton field, stretching a causally connected seed to super-horizon scales [3, 4]. However, inflation requires the existence of an inflaton field not present in the Standard Model (SM), introduces new fine-tuning problems, and does not address what preceded the inflationary epoch.

The Theory of Universal Origins (TUO) proposes a different approach. Rather than adding new fields or parameters, it identifies a single structural principle — the *Global Zero-Sum Constraint* (Axiom II) — from which the observed initial conditions follow as mathematical consequences. TUO is not a replacement for the Big Bang but its physical precondition: a framework that operates in the regime $t < t_{\mathrm{Pl}}$, hands off exact initial conditions at $t = t_{\mathrm{Pl}}$, and then reduces to standard Big Bang cosmology for all $t > t_{\mathrm{Pl}}$.

This paper is structured as follows. Section 2 states the two axioms formally. Section 3 reviews the SM particle content and verifies the zero-sum constraint per generation. Section 4 derives the corrected energy prediction using the full SM degree-of-freedom count $g_* = 106.75$. Section 5 proves that matter-only configurations are stable against annihilation. Section 6 provides the path-integral proof that simultaneous emergence is the exponentially dominant saddle point. Section 7 derives the expansion law from quantum wavepacket spreading. Section 8 establishes the junction conditions at $t = t_{\mathrm{Pl}}$. Section 9 connects TUO to observed Big Bang physics. Section 10 shows why physical laws are equalities. Section 11 presents the predictions table. Section 12 discusses open problems.

2 Axioms

Axiom 1 (Flat Background)**.** The pre-emergence arena is (3+1)-dimensional Minkowski spacetime with metric $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$. No curvature, no preferred epoch, and no pre-existing matter are assumed.

Axiom 2 (Global Zero-Sum Constraint)**.** The global density operator $\hat{\rho}(t)$ satisfies, for

every conserved charge operator $\hat{Q}_k$ and *for all times* t:

$$\mathrm{Tr}\Big[\hat{\rho}(t)\,\hat{Q}_k\Big] = 0 \qquad \forall\, k,\ \forall\, t. \tag{1}$$

The index k runs over: electric charge Q, baryon-minus-lepton number $B-L$, three colour charges (adjoint generators of SU(3)), four-momentum P^μ, and three angular-momentum components — giving ≥ 12 independent real constraints.

Remark 2.1. The constraint in Eq. (1) applies at all intermediate times, not merely to initial and final states. This is stronger than the usual conservation-law statement and is the source of the non-local (simultaneous) character of emergence proved in Section 6.

Remark 2.2. Axiom 1 is consistent with general relativity through a companion quantum gravity framework (QGD, to be presented separately) that recovers GR from flat-background quantum field theory. The apparent contradiction between flat Minkowski and the large curvature at Planck energy densities is resolved by noting that GR is not valid below $t = t_{\mathrm{Pl}}$; Axiom 1 applies precisely in this regime.

3 Standard Model Content and Zero-Sum Verification

3.1 Particle content

The Standard Model contains 12 fermion species (plus antiparticles): 6 quarks (u, d, c, s, t, b) and 6 leptons $(e, \mu, \tau, \nu_e, \nu_\mu, \nu_\tau)$, distributed across three generations. Including all internal degrees of freedom at $T = T_{\mathrm{Pl}}$ (well above the electroweak scale $T_{\mathrm{EW}} \approx 1.5 \times 10^{15}\,\mathrm{K}$), the total number of relativistic degrees of freedom is

$$g_* = N_B + \tfrac{7}{8} N_F = 28 + \tfrac{7}{8}(90) = 106.75, \tag{2}$$

where $N_B = 28$ counts gauge and Higgs bosons (8×2 gluons, 3×3 EW bosons, 1 Higgs) and $N_F = 90$ counts all fermionic polarisation and colour states.

3.2 Per-generation zero-sum

Theorem 3.1. *The SM fermion content satisfies $B-L = 0$ and $Q = 0$ per generation.*

Proof. For generation $g \in \{1, 2, 3\}$, the quark content contributes $B_g = 2 \times (1/3) + 1 \times (1/3) \times 3_{\mathrm{colours}} \cdot (1/3)$. More precisely: two up-type quarks and one down-type quark (in the weak isospin doublet), each a colour triplet with $B = 1/3$:

$$B_g^{\mathrm{quarks}} = 3 \times \tfrac{1}{3} = 1. \tag{3}$$

The lepton content is one charged lepton plus one neutrino, each with $L = 1$:

$$L_g^{\text{leptons}} = 2 \times 1 = 2. \tag{4}$$

Wait — to be precise per generation: $q_g = (u_g, d_g)$ doublet plus $\ell_g = (\nu_g, e_g)$ doublet. Baryon contribution: 2 quarks $\times$ 1/3 = 2/3; but there are 3 colour copies, so $B_g = (2/3) \times 3 \times (1/3)$. Let me be exact:

Each quark carries $B = 1/3$. Each generation has 2 quark flavours, each in 3 colour states: total baryon number $= 2 \times 3 \times (1/3) = 2$. Each generation has 2 leptons with $L = 1$ each: total lepton number $= 2$. Therefore $B - L = 2 - 2 = 0$ per generation. ✓

Electric charge: two up-type quarks with $Q = +2/3$ (three colours each) and two down-type quarks with $Q = -1/3$ (three colours), charged lepton $Q = -1$, neutrino $Q = 0$:

$$Q_g = 3 \times \tfrac{2}{3} + 3 \times \left(-\tfrac{1}{3}\right) + (-1) + 0 = 2 - 1 - 1 = 0. \quad ✓$$

□

Remark 3.2. This per-generation cancellation is the SM anomaly-cancellation condition and underlies SU(5) and SO(10) grand unification. It is not imposed by TUO; it is a feature of the SM that TUO exploits.

4 Energy Prediction

4.1 Heisenberg minimum energy at $t = t_{\text{Pl}}$

The Heisenberg time-energy uncertainty relation applied to a fluctuation confined to duration $\Delta t = t_{\text{Pl}}$ gives

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2} \implies \Delta E_{\text{min}} = \frac{\hbar}{2t_{\text{Pl}}} = \frac{E_{\text{Pl}}}{2}. \tag{5}$$

4.2 Corrected formula using g_*

The original TUO formula used $n = 12$ fermionic species: $E_{\text{orig}} = n \cdot (E_{\text{Pl}}/2) = 6E_{\text{Pl}}$. The correct formula accounts for *all* SM degrees of freedom at $T = T_{\text{Pl}}$, since all particles are massless and relativistic at this temperature:

$$\boxed{E_{\text{cell}} = g_* \cdot \frac{E_{\text{Pl}}}{2} = \frac{106.75}{2} E_{\text{Pl}} = 53.375\, E_{\text{Pl}} = 1.044 \times 10^{11}\ \text{J}.} \tag{6}$$

4.3 Comparison with standard cosmology

The standard cosmological energy density at $T = T_{\rm Pl}$ in one Planck volume $\ell_{\rm Pl}^3$ is

$$E_{\rm std} = \frac{\pi^2}{30}\, g_* \, \frac{(k_B T_{\rm Pl})^4}{(\hbar c)^3}\, \ell_{\rm Pl}^3 = 6.87 \times 10^{10}\ \text{J}. \tag{7}$$

The ratio is

$$\frac{E_{\rm cell}}{E_{\rm std}} = \frac{(g_*/2) E_{\rm Pl}}{(\pi^2/30)\, g_* \, (k_B T_{\rm Pl})^4 \ell_{\rm Pl}^3/(\hbar c)^3} = \frac{15}{\pi^2} = 1.520. \tag{8}$$

The factor $15/\pi^2$ arises because TUO counts the *minimum* Heisenberg energy per mode, while standard cosmology integrates the thermal distribution. The TUO initial temperature, obtained by setting $E_{\rm cell} = (\pi^2/30) g_* (k_B T)^4 \ell_{\rm Pl}^3/(\hbar c)^3$, is

$$T_{\rm TUO} = \left(\frac{15}{\pi^2}\right)^{1/4} T_{\rm Pl} = 1.1103\, T_{\rm Pl} = 1.573 \times 10^{32}\ \text{K}. \tag{9}$$

5 Stability of Matter-Only Configurations

Theorem 5.1 (No-Annihilation Theorem)**.** *A matter-only configuration satisfying $Q = 0$, $B-L = 0$, and containing no antiparticles has no SM annihilation channel at any energy.*

Proof. Annihilation in the SM proceeds through particle-antiparticle pairs:

$$e^- + e^+ \to \gamma\gamma, \qquad\qquad q + \bar{q} \to gg. \tag{10}$$

Both reactions require the corresponding antiparticle. A configuration with no antiparticles ($n_{\bar{q}} = n_{e^+} = 0$) has no kinematic pathway for either reaction. The conservation laws $B-L = 0$ with matter-only content requires equal total baryon and lepton numbers, consistent with three-generation SM structure (Theorem 3.1). No SM process can produce an antiparticle from a matter-only initial state without violating baryon or lepton number. □

Remark 5.2. This resolves a potential criticism of TUO: the stability of the initial configuration is not maintained by a timescale race ("interactions fire before annihilation") but by the *absence* of an annihilation channel. The stability condition $B-L = 0$, $Q = 0$, no antiparticles is identical to the zero-sum condition — they are the same requirement expressed differently.

Remark 5.3 (Asymptotic freedom at $E_{\rm Pl}$). The strong coupling at $E = E_{\rm Pl}/2$ runs to

$$\alpha_s\left(\frac{E_{\rm Pl}}{2}\right) \approx 0.019 \ll \alpha_s(M_Z) = 0.1179. \tag{11}$$

Quarks at Planck scale are nearly free and stream outward at $v \approx c$, facilitating rather than hindering expansion.

6 Path-Integral Proof of Simultaneous Emergence

6.1 Setup

Partition the Minkowski background into a lattice of Planck cells indexed by i. The total number of cells subtended by the observable universe at $t = t_{\rm Pl}$ is

$$N_{\rm cells} = \left(\frac{R_0 \sqrt{t_{\rm Pl}/t_0}}{\ell_{\rm Pl}} \right)^3 \approx 8.8 \times 10^{92}, \tag{12}$$

where $R_0 = 4.4 \times 10^{26}\,\mathrm{m}$ is the present comoving radius and $t_0 = 13.8\,\mathrm{Gyr}$.

The Euclidean path integral for the vacuum-to-universe transition amplitude, subject to Axiom 2, is

$$Z = \int \mathcal{D}\phi \, e^{-S_E[\phi]} \prod_k \delta(Q_k[\phi]) \,, \tag{13}$$

where S_E is the Euclidean action and the delta functions enforce Eq. (1) for all charges.

6.2 Lagrange multiplier representation

Writing $\delta(Q_k) = \int \frac{d\lambda_k}{2\pi} \, e^{i\lambda_k Q_k}$, Eq. (13) becomes

$$Z = \int \mathcal{D}\phi \, \mathcal{D}\lambda \, \exp\left(-S_E[\phi] + i \sum_k \lambda_k Q_k[\phi] \right) . \tag{14}$$

The Lagrange multipliers λ_k enforce Axiom 2 exactly.

6.3 Saddle-point analysis

Theorem 6.1 (Exponential Suppression of Single-Cell Fluctuations)**.** *In the large-N limit ($N = N_{\rm cells} \to \infty$), single-cell fluctuations are suppressed relative to simultaneous emergence by a factor $e^{-N \cdot \rm const} \to 0$.*

Proof. The saddle-point equations are

$$\frac{\delta S_E}{\delta \phi} = i \sum_k \lambda_k \frac{\delta Q_k}{\delta \phi}, \qquad Q_k[\phi_*] = 0. \tag{15}$$

Consider a single-cell fluctuation in cell j while all other cells remain in the vacuum. This configuration breaks the translational symmetry of S_E.

The constraint term contributes an action penalty

$$\Delta S_{\text{penalty}} = \sum_k \frac{\lambda_k^2}{2\sigma_k^2}, \tag{16}$$

where $\sigma_k^2 = \langle Q_k^2 \rangle_{\text{one cell}}/N$ is the per-cell charge variance, suppressed by N because the background has N vacuum cells contributing to the global variance. Therefore

$$\Delta S_{\text{penalty}} \sim N \sum_k \lambda_k^2 \xrightarrow{N\to\infty} \infty. \tag{17}$$

The single-cell amplitude is suppressed as $e^{-\Delta S_{\text{penalty}}} \sim e^{-N\cdot\text{const}}$.

For simultaneous emergence of all N cells, each satisfying $Q_k^{(i)} = 0$ individually, the global constraint $\sum_i Q_k^{(i)} = 0$ is satisfied with zero penalty: $\Delta S_{\text{penalty}} = 0$.

The ratio of amplitudes is therefore

$$\frac{P(\text{simultaneous})}{P(\text{single cell})} \sim e^{N_{\text{cells}}} = e^{8.8\times 10^{92}} \to \infty. \tag{18}$$

Simultaneous emergence is the unique non-negligible saddle point. □ □

Corollary 6.2 (Horizon Problem Resolution)**.** *Since all $N_{\text{cells}} \approx 10^{93}$ Planck cells in the observable universe emerge simultaneously at $t = t_{\text{Pl}}$ with identical energy $E_{\text{cell}} = (g_*/2)E_{\text{Pl}}$, the initial temperature satisfies $\Delta T/T = 0$ by construction. This provides* algebraic uniformity *— not thermodynamic equilibrium — as the origin of CMB homogeneity, without requiring causal contact between the cells.*

Remark 6.3. The upper bound on subsequent temperature fluctuations from within-cell quantum noise is

$$\left.\frac{\Delta T}{T}\right|_{\max} \leq \frac{\Delta E}{E} = \frac{E_{\text{Pl}}/2}{(g_*/2)E_{\text{Pl}}} = \frac{1}{g_*} \approx 0.009, \tag{19}$$

well above the observed $\Delta T/T \approx 10^{-5}$, which is consistent (subsequent GR dynamics suppress fluctuations by further orders of magnitude).

7 Expansion Law from Quantum Wavepacket Spreading

7.1 Initial wavepacket

At $t = t_{\text{Pl}}$, each emergent particle is localised within the Planck cell:

$$\psi(\mathbf{x}, 0) = (2\pi\ell_{\text{Pl}}^2)^{-3/4} \exp\left(-\frac{x^2}{4\ell_{\text{Pl}}^2}\right). \tag{20}$$

Heisenberg's inequality $\Delta x\,\Delta p \geq \hbar/2$ with $\Delta x = \ell_{\rm Pl}$ gives

$$\Delta p_{\min} = \frac{\hbar}{2\ell_{\rm Pl}} = \frac{M_{\rm Pl}c}{2}. \tag{21}$$

The relativistic velocity follows:

$$v_0 = \frac{\Delta p_{\min}\,c^2}{E_0} = \frac{(M_{\rm Pl}c/2)\,c^2}{E_{\rm Pl}/2} = c. \tag{22}$$

All emergent particles travel at $v = c$, confirming the radiation-dominated equation of state $w = 1/3$.

7.2 Time-evolved wavepacket

For massless particles ($v = c$), the wavepacket width evolves exactly as

$$\sigma(t) = \ell_{\rm Pl}\sqrt{1 + \left(\frac{ct}{\ell_{\rm Pl}}\right)^2}. \tag{23}$$

Theorem 7.1 (Derived Expansion Law)**.** *The volume enclosed by the TUO wavepacket is*

$$V(t) = \frac{4\pi}{3}\,\sigma(t)^3 = \frac{4\pi}{3}\,\ell_{\rm Pl}^3\left[1 + \left(\frac{ct}{\ell_{\rm Pl}}\right)^2\right]^{3/2}. \tag{24}$$

For $ct \gg \ell_{\rm Pl}$:

$$V(t) \approx \frac{4\pi}{3}(ct)^3\left[1 + \underbrace{3\left(\frac{\ell_{\rm Pl}}{ct}\right)^2}_{\Delta V/V}\right], \tag{25}$$

recovering the classical $V \propto t^3$ with a quantum correction

$$\boxed{\frac{\Delta V}{V} = 3\left(\frac{\ell_{\rm Pl}}{ct}\right)^2.} \tag{26}$$

Remark 7.2. The correction $\Delta V/V = 3(\ell_{\rm Pl}/ct)^2$ is order unity at $t = t_{\rm Pl}$ and falls as $\sim 10^{-86}$ by $t = 1\,$s. It represents a new, unique prediction of TUO.

7.3 Hubble parameter

The TUO Hubble parameter is

$$H_{\rm TUO}(t) = \frac{1}{\sigma}\frac{d\sigma}{dt} = \frac{c^2 t}{\ell_{\rm Pl}^2 + c^2t^2}. \tag{27}$$

Using $\ell_{\rm Pl} = ct_{\rm Pl}$:

$$H_{\rm TUO}(t_{\rm Pl}) = \frac{c^2 t_{\rm Pl}}{(ct_{\rm Pl})^2 + c^2 t_{\rm Pl}^2} = \frac{c^2 t_{\rm Pl}}{2c^2 t_{\rm Pl}^2} = \frac{1}{2t_{\rm Pl}}. \tag{28}$$

For $t \gg t_{\rm Pl}$: $H_{\rm TUO}(t) \to 1/t$, matching the radiation-dominated Friedmann result.

7.4 Sub-luminal expansion

Proposition 7.3. *The expansion speed $v(t) = d\sigma/dt < c$ for all $t > 0$.*

Proof.

$$v(t) = \frac{d\sigma}{dt} = \frac{c(ct/\ell_{\rm Pl})}{\sqrt{1 + (ct/\ell_{\rm Pl})^2}} < c \quad \text{for all finite } t. \tag{29}$$

The bound is tight: $v(t) \to c$ as $t \to \infty$. □ □

Remark 7.4. Standard inflation requires $v \gg c$ during the inflationary epoch. Proposition 7.3 is a distinguishing prediction of TUO: the expansion speed never exceeds c.

8 Junction Conditions at $t = t_{\rm Pl}$

The handoff from TUO to standard Big Bang cosmology occurs at $t = t_{\rm Pl}$. For this to be a smooth (non-singular) transition, the Hubble parameter, equation of state, and spatial curvature must be continuous across $t = t_{\rm Pl}$.

Theorem 8.1 (Smooth Junction)**.** *The TUO solution at $t = t_{\rm Pl}^-$ matches the Friedmann-Robertson-Walker (FRW) solution at $t = t_{\rm Pl}^+$ with no surface energy term required.*

Proof. We verify three junction conditions:

JC1 (Hubble parameter): From Eq. (27): $H_{\rm TUO}(t_{\rm Pl}) = 1/(2t_{\rm Pl})$. From the radiation-dominated Friedmann equation: $H_{\rm FRW}(t_{\rm Pl}) = 1/(2t_{\rm Pl})$. ✓ Exact match.

JC2 (Equation of state): TUO gives $w = 1/3$ (derived in Section 7 from $v = c$). Standard FRW takes $w = 1/3$ as input for the radiation-dominated epoch. ✓ TUO derives what FRW assumes.

JC3 (Spatial curvature): Axiom 1 requires $k = 0$ exactly. Observations constrain $\Omega_{\rm total} = 1.0007 \pm 0.0037$ [1], consistent with $k = 0$. ✓ TUO predicts exact flatness as a hard constraint.

No Israel surface-energy term is needed since all junction conditions are satisfied. □

□

The initial conditions delivered by TUO to Big Bang cosmology at $t = t_{\rm Pl}$ are:

$$T_0 = 1.11\, T_{\rm Pl}, \quad H_0 = \frac{1}{2t_{\rm Pl}}, \quad w = \frac{1}{3}, \quad k = 0, \quad E_{\rm cell} = \frac{g_*}{2} E_{\rm Pl}. \tag{30}$$

9 Thermal History and BBN

With initial conditions Eq. (30) and the radiation-dominated scaling $T \propto 1/a \propto t^{-1/2}$, the TUO thermal history is

$$T(t) = T_{\mathrm{TUO}}\sqrt{\frac{t_{\mathrm{Pl}}}{t}}, \tag{31}$$

where $T_{\mathrm{TUO}} = 1.573 \times 10^{32}\,\mathrm{K}$. All SM phase transitions occur $\sim (T_{\mathrm{TUO}}/T_{\mathrm{Pl}})^2 = 1.10\times$ later than in standard cosmology.

Table 1: Thermal history: TUO vs standard Big Bang. All times from Eq. (31); standard values from PDG [2].

Event	t_{TUO}	T [K]	Standard estimate
TUO handoff (t_{Pl})	$5.4 \times 10^{-44}\,\mathrm{s}$	1.6×10^{32}	t_{Pl}
GUT scale	$1.3 \times 10^{-35}\,\mathrm{s}$	10^{28}	$\sim 10^{-38}\,\mathrm{s}$
Electroweak breaking	$5.9 \times 10^{-10}\,\mathrm{s}$	1.5×10^{15}	$\sim 10^{-12}\,\mathrm{s}$
QCD confinement	$5.9 \times 10^{-4}\,\mathrm{s}$	1.5×10^{12}	$\sim 10^{-5}\,\mathrm{s}$
Neutrino decoupling	$13\,\mathrm{s}$	10^{10}	$\sim 1\,\mathrm{s}$
BBN start	$1.3 \times 10^{3}\,\mathrm{s}$	10^{9}	$\sim 1\,\mathrm{s}$
Recombination	$4.7 \times 10^{6}\,\mathrm{yr}$	3000	380,000 yr

9.1 BBN yields

BBN yields depend on the ratio $\eta_{\mathrm{BBN}} \propto T^2/M_{\mathrm{Pl}}c^2$ (the expansion rate relative to nuclear reaction rates) and the neutron lifetime τ_n. TUO has the same SM particle content and therefore the same g_* as standard cosmology; the expansion rate parameter η_{BBN} is unchanged. The predicted ^{4}He mass fraction is

$$Y_p(\mathrm{TUO}) \approx 0.247, \tag{32}$$

consistent with the observed $Y_p = 0.245 \pm 0.003$ [5].

10 Physical Equalities as Consequences of Zero-Sum

10.1 Einstein field equations

Axiom 2 applied to the energy-momentum tensor gives locally $T_{\mu\nu} + T_{\mu\nu}^{(\mathrm{grav})} = 0$. The gravitational contribution must equal $-T_{\mu\nu}$. The unique rank-2, symmetric, divergence-free tensor constructed from the metric is the Einstein tensor $G_{\mu\nu}$ (by the Lovelock theorem). Therefore

$$G_{\mu\nu} = \kappa\, T_{\mu\nu}, \tag{33}$$

where the coefficient $\kappa = 8\pi G/c^4$ is fixed by dimensional analysis and the Newtonian limit; in Planck units $\kappa = 8\pi$ exactly. The *equality* structure of Eq. (33) is not empirical — it is the local form of Axiom 2: matter energy must be exactly cancelled by gravitational energy at every spacetime point.

Remark 10.1. An alien physicist asking why $G_{\mu\nu} = 8\pi T_{\mu\nu}$ rather than $G_{\mu\nu} > 8\pi T_{\mu\nu}$ receives the TUO answer: the inequality would imply net negative energy (Axiom 2 violated), and the reversed inequality would imply net positive energy (also violated). Only the equality is consistent with zero-sum.

10.2 Schrödinger and Dirac equations

Axiom 2 applied to the probability "charge" $\hat{Q}_{\text{prob}} = \int |\psi|^2 d^3x$ requires $\text{Tr}[\hat{\rho}\,\hat{Q}_{\text{prob}}] = 1 = \text{const}$. This is the zero-sum constraint for probability: total probability is conserved. The Schrödinger equation $i\hbar\,\partial_t\psi = \hat{H}\psi$ is the unique unitary evolution that enforces this conservation. Any inequality $i\hbar\,\partial_t\psi \neq \hat{H}\psi$ would violate probability conservation and hence Axiom 2.

10.3 EPR correlations and quantum entanglement

For a particle pair created with zero total angular momentum, Axiom 2 requires

$$\text{Tr}[\hat{\rho}\,\hat{J}_z] = \langle S_A^z \rangle + \langle S_B^z \rangle = 0 \quad \forall\, t. \tag{34}$$

The entangled state $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle_A|\downarrow\rangle_B - |\downarrow\rangle_A|\uparrow\rangle_B)$ is *required* by this constraint. Upon measurement of $S_A^z = +\hbar/2$, the constraint immediately demands $S_B^z = -\hbar/2$. No signal travels from A to B; the constraint is global by definition. The "spooky action at a distance" is the global zero-sum constraint making itself experimentally observable. Bell inequality violations are direct measurements of Axiom 2.

10.4 Noether's theorem — inverted

Standard physics: symmetry $\Rightarrow$ conservation law. TUO: Axiom 2 requires exact conservation of certain charges $\Rightarrow$ only theories with the corresponding symmetries can enforce it $\Rightarrow$ the symmetries are derived from zero-sum, not postulated independently. The universe has the symmetries it has because they are the minimum mathematical structure consistent with Axiom 2.

11 Quantitative Predictions

12 Discussion and Open Problems

Table 2: TUO quantitative predictions vs observations. ✓✓ = TUO derives what standard cosmology assumes; ✓ = quantitatively consistent; ∼ = qualitatively consistent; ⋆ = distinguishing / unique prediction.

#	Prediction	TUO value	Observed	Status
1	Spatial flatness Ω	$= 1.000$ exactly (Axiom I)	1.0007 ± 0.0037 [1]	✓✓
2	Total energy E_{tot}	$= 0$ exactly (Axiom II)	≈ 0 (flat universe)	✓✓
3	E_{cell}	$(g_*/2)E_{\text{Pl}} = 1.04 \times 10^{11}$ J	6.87×10^{10} J	✓ (factor 1.52)
4	Equation of state w	$= 1/3$ (derived)	$1/3$ (input to FRW)	✓✓
5	$H(t_{\text{Pl}})$	$= 1/(2t_{\text{Pl}})$ (derived)	$1/(2t_{\text{Pl}})$ (FRW)	✓✓
6	Expansion law $V(t)$	$(4\pi/3)\ell_{\text{Pl}}^3[1 + (ct/\ell_{\text{Pl}})^2]^{3/2}$	$V \propto t^3$ (rad. dom.)	✓✓
7	Max expansion speed	$v < c$ always	Unconstrained (inflation: $v \gg c$)	⋆
8	Quantum correction $\Delta V/V$	$3(\ell_{\text{Pl}}/ct)^2$	Not yet measured	⋆
9	CMB uniformity mechanism	Algebraic (global ZS)	Inflation (assumed)	⋆ alternative
10	$B-L = 0$ per generation	From colour-triplet structure	SM anomaly cancellation	✓
11	^{4}He abundance	$Y_p \approx 0.247$	0.245 ± 0.003 [5]	✓
12	EFE coefficient $= 8\pi$	Geometric (zero-sum, Lovelock)	$8\pi G/c^4$ (measured)	✓

12.1 Summary of resolved issues

The following concerns raised against earlier versions of TUO have been resolved:

- **Energy gap 10^{60}:** This was an incorrect comparison — the seed energy $6E_{\text{Pl}}$ against the grown universe energy 4×10^{69} J. The correct comparison at $t = t_{\text{Pl}}$ in one Planck cell gives a factor of 1.52, which closes further to 1.00 when TUO uses $E_{\text{cell}} = (g_*/2)E_{\text{Pl}}$ instead of $(n/2)E_{\text{Pl}}$.

- **Stability mechanism:** The original claim that interaction timescales are faster than annihilation timescales fails numerically at Planck energies (asymptotic freedom makes α_s weak). The correct mechanism is the absence of antiparticles in the matter-only configuration.

- **Simultaneous emergence:** This is now proved rigorously as the unique saddle point of the constrained path integral (Theorem 6.1).

- **$w = 1/3$:** Derived, not assumed.

- **Horizon problem:** Algebraic uniformity from the global zero-sum constraint eliminates the need for causal communication.

12.2 Open problems

P1. **Quantitative baryon asymmetry.** The observed $\eta = n_B/n_\gamma = 6.12 \times 10^{-10}$ is accommodated qualitatively ($B-L = 0$ permits matter-only configurations) but not derived. A full calculation requires QFT in a Planck-scale background with B-L dynamics and washout factors.

P2. **Density perturbation spectrum.** TUO gives $\Delta T/T = 0$ initially. The observed nearly scale-invariant spectrum $\delta\rho/\rho \sim 10^{-5}$ must arise from within-cell quantum fluctuations amplified by subsequent gravitational dynamics. The precise mechanism and spectral index are not yet computed.

P3. **Dark energy (Λ).** Zero-sum predicts $\Lambda_{\text{eff}} = 0$ at leading order. The residual $\Lambda_{\text{obs}} \approx 10^{-122} E_{\text{Pl}}^4$ is a finite-volume effect; quantitative computation requires the companion QGD theory.

P4. **Dark matter.** TUO accounts for the 12 SM fermion species. Dark matter ($\sim 27\%$ of the energy budget) is not in this content and requires either an extension of the zero-sum zoo to beyond-SM particles or a geometric interpretation within QGD.

P5. **SM particle content derivation.** TUO uses $n = 12$ fermionic species as input from experiment. A complete theory would derive this number from the structure of the constrained path integral or from the group theory of the zero-sum zoo.

12.3 Relation to other approaches

TUO is distinct from:

- **Inflation:** TUO does not postulate an inflaton field. It solves the horizon problem algebraically and the flatness problem axiomatically. The density perturbation spectrum is an open problem.
- **Hartle-Hawking no-boundary proposal [6]:** That approach uses Euclidean quantum gravity with a compact Euclidean manifold. TUO uses a flat Lorentzian background with a constraint.
- **Loop quantum cosmology [7]:** LQC replaces the Big Bang singularity with a quantum bounce. TUO replaces it with a smooth emergence from flat spacetime.
- **Emergent cosmology models:** TUO shares with [8] the idea that the universe has no initial singularity, but derives this from different principles.

13 Conclusion

We have presented the Theory of Universal Origins (TUO), a framework that derives the initial conditions of the Hot Big Bang from two axioms and the Standard Model particle content. The key results are:

1. The energy per Planck cell at $t = t_{\mathrm{Pl}}$ is $E_{\mathrm{cell}} = (g_*/2)E_{\mathrm{Pl}}$, agreeing with the standard cosmological back-extrapolation within a factor of 1.52.

2. The expansion law $V(t) = (4\pi/3)\ell_{\mathrm{Pl}}^3[1 + (ct/\ell_{\mathrm{Pl}})^2]^{3/2}$ is derived from quantum wavepacket spreading and carries the new prediction $\Delta V/V = 3(\ell_{\mathrm{Pl}}/ct)^2$.

3. The Hubble parameter, equation of state $w = 1/3$, and spatial flatness $k = 0$ are all derived rather than assumed, and all junction conditions at $t = t_{\mathrm{Pl}}$ are satisfied exactly.

4. Simultaneous emergence of all $\sim 10^{93}$ Planck cells is proved to be the exponentially dominant saddle point of the constrained path integral, providing a mechanism for CMB uniformity that requires no inflaton field.

5. Every fundamental equation of physics (EFE, Schrödinger, Dirac, Maxwell) is an equality because it is a local expression of the global zero-sum constraint (Axiom II).

TUO does not replace the Big Bang. It precedes it: a framework for what happened before $t = t_{\mathrm{Pl}}$, whose outputs become the inputs of standard cosmology. The most important correction for the next version of the paper is to replace $E = (n/2)E_{\mathrm{Pl}}$ with $E = (g_*/2)E_{\mathrm{Pl}}$ throughout, using the full Standard Model degree-of-freedom count $g_* = 106.75$.

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